

EXTREME DISTRIBUTION FOR A GENERALIZED STOCHASTIC VOLATILITY MODEL¹

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Summary: We study the asymptotic behaviour of the extreme values of a stochastic volatility model (Y_t) when the noise follows a generalized error distribution (GED), see Box and Tiao (1973). In this paper, we show that in a general context, the normalized extremes of a log-transformation of (Y_t) converge in distribution to the double exponential distribution. We investigate the importance of the different assumptions using Monte Carlo simulations. We also deal with the finite sample behaviour of the normalized maxima. The influence of the parameters of the model is discussed.

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1. Introduction

The class of stochastic volatility models has been extensively examined in finance and financial econometrics. Indeed, volatility plays a central role in the analysis of a lot of phenomena in these domains. Here we are interested by a discrete time version of the volatility model introduced first by Taylor (1986). This model appears as a particular model of the Stochastic Autoregressive Volatility (SARV) model introduced by Andersen in 1994. The univariate volatility model (Y_t) that we consider, can be defined by the following equation:

$$Y_t = \sigma \exp\left(\frac{\alpha_t}{2}\right) \varepsilon_t \quad (1.1)$$

with

$$\alpha_t = \sum_{j=0}^{\infty} \theta_j Z_{t-j}, \quad (1.2)$$

where σ is a positive constant, (ε_t) is a sequence of independent and identically distributed (iid) random variables (r.v.) and (Z_t) a sequence of iid Gaussian r.v. with mean zero, variance σ_Z^2 . This last series is independent of the sequence (ε_t) . The parameters (θ_j) are such that $\sum_{j \geq 0} |\theta_j|^2 < \infty$. Generally, in the literature, it is assumed that the r.v. ε_t and Z_t are not correlated for all t . For simplicity here we will assume independence between the two sequences.

The model (1.1) picks some asymmetry behaviour that is often found in stock prices and a negative correlation between ε_t and Z_t induces a leverage effect. This explains why practitioners often use this model. The behaviour of the autocorrelation function and the power-moments for the process defined by (1.1)–(1.2) are well known when (ε_t) is Gaussian or Student, see for instance Harvey (1993), Taylor (1986), Ghysels *et al.* (1996) or Shepard (1996).

However, returns on financial assets such as stocks often exhibit excess kurtosis in the marginal distribution. To allow for the possibility of non-normality in the conditional distribution of returns, many authors assume that the noise is iid and drawn from the so-called Generalized Error Distribution (GED). First, Nelson (1991) proposed a new form of Autoregressive Conditional Heteroskedasticity (ARCH) model with GED errors applied to CRSP value-weighted Market Index from 1962 to 1987. Furthermore, Anderson (1994) found evidence that the continuously compounded daily returns on the IBM common stock over the period January 1, 1973 to December 31, 1991 are slightly skewed to the left and have excess kurtosis relative to the normal distribution. He suggested a Stochastic Autoregressive Volatility model with GED errors (SARV-GED). Taylor (1994) compared several volatility models with the three most popular conditional distributions : the normal, the scaled t and the GED, using exchange rates. He found that conditional normal distributions are unacceptable. He also proved that the GED is superior to the t -distribution for describing the conditional distribution of these exchange rates, for the period studied.

Recently banks and insurance companies have been faced with questions concerning rare events. In insurance, these extremal events may clearly correspond to individual claims which by far exceed the capacity of a single insurance company. In finance, these extremal events are present, in particular, during major stock market crashes, for instance during the 1987 crisis. One of the first worries of these institutions is to define a well-functioning risk management and a control system to address these problems. Here, we propose specific tools to try to treat these problems.

In this context, it is important to know the extremal distribution of the processes which characterize the behaviour of asset prices. Our

contribution consists in studying the distribution's extremal behaviour for some transformation of the process (Y_t) defined by (1.1)–(1.2). We assume that the noise (ε_t) follows a GED. This class of distributions has been studied in Box and Tiao (1973), for instance. The distribution's extremal behaviour for the process (Y_t) defined by (1.1)–(1.2) has already been studied by Breidt and Davis (1998) when (ε_t) is Gaussian. Here we extend this result and show that the normalized extremes of a log-transformation of (Y_t) converge in distribution to the double exponential distribution when the distribution of (ε_t) is GED. Since we are interested in a feasible application of these results on real data, we investigate the importance of the different assumptions using Monte Carlo simulations. Thus, we show that these assumptions are not quite relevant in the reality context. In fact, the results obtained in an asymptotic context performed badly in a finite context. The latter situation is well known in extreme value theory. Obviously, this implies some difficulties to use these results directly to construct a risk management theory.

Our paper is organized as follows: the extremal behaviour of a log-transformation of (Y_t) is presented in Section 2, in Section 3 we focus on finite sample behaviour for the normalized maxima of the log-transformation of the process (Y_t) . Finally, Section 4 is devoted to the conclusion and the proofs are contained in an Appendix.

2. Asymptotic behaviour of the maxima of the process (X_t)

In this section we study the distribution's asymptotic behaviour of a class of transformations for the process (Y_t) defined in (1.1)–(1.2), when (ε_t) follows a Generalized Error Distribution (GED). The GED's density is defined by

$$f_\varepsilon(x) = c_0 \exp(-k|x|^\gamma), \quad (2.1)$$

with $c_0 > 0$, $\gamma > 0$ and $k > 0$. The expression (2.1) can also be rewritten as

$$f_\varepsilon(x) = \frac{\gamma \exp(-\frac{1}{2}|\frac{x}{\tau}|^\gamma)}{\tau 2^{1+\frac{1}{\gamma}} \Gamma(\frac{1}{\gamma})}, \quad \gamma > 0 \quad (2.2)$$

with

$$\tau = \left[\frac{2^{-\frac{2}{\gamma}} \Gamma(\frac{1}{\gamma})}{\Gamma(\frac{3}{\gamma})} \right]^{\frac{1}{2}}.$$

This class of densities contains the Gaussian density ($\gamma = 2$) and the Laplace density ($\gamma = 1$), and has the uniform density as a limit ($\gamma \rightarrow^o +\infty$). It was first introduced by Subbotin (1923) as the exponential power distribution. The tail behaviour of the innovations process (ε_t) , characterized by such a density, depends on the tail-thickness parameter γ . For instance, if $\gamma = 2$, then $\varepsilon_t \sim \mathcal{N}(0, 1)$, while for $\gamma < 2$ the distribution has thicker tails than the Gaussian distribution.

Now we define for all t ,

$$X_t = \ln\left(\frac{Y_t}{\sigma}\right)^2 = \alpha_t + \ln \varepsilon_t^2. \quad (2.3)$$

and we put

$$\zeta_t = \ln \varepsilon_t^2 \text{ for all } t \in \mathbb{Z}, \quad (2.4)$$

thus the process (X_t) becomes a sum of two independent processes :

$$X_t = \alpha_t + \zeta_t. \quad (2.5)$$

The model (2.5) can be considered as a regression model with a particular noise (ζ_t) . When (ε_t) is characterized by a GED, we denote by F the distribution function of (X_t) . First of all, we study the asymptotic behaviour of the distribution function F .

Proposition 1. *Let (X_t) be the process defined by (2.5) and assume that the distribution of (ε_t) is defined in (2.1), then the asymptotic behaviour of F is*

$$\bar{F}(x) = \mathbb{P}\{X_t > x\}$$

$$\sim \frac{\sigma_\alpha}{\sqrt{\pi}} \exp \left\{ -\frac{x^2}{2\sigma_\alpha^2} + \frac{ax \ln g(x)}{\sigma_\alpha^2} - \frac{2x(1+R)}{\gamma\sigma_\alpha^2} + A \ln g(x) - \frac{bx \ln g(x)}{\sigma_\alpha^2 g(x)} \right. \\ \left. - \frac{2 \ln^2 g(x)}{\sigma_\alpha^2 \gamma^2} - \frac{cx}{\sigma_\alpha^2 g(x)} + \frac{1}{8\sigma_\alpha^2} \left(\frac{1}{2} \gamma - 1 \right)^2 + \frac{1}{\sigma_\alpha^2} \left(\frac{1}{2} - \frac{1}{\gamma} \right) + C + \mathcal{O} \left(\frac{\ln^2 g(x)}{g(x)} \right) \right\}, \quad (2.6)$$

where

$$a = \frac{2}{\gamma}, \quad b = \left(\frac{2}{\gamma} \right)^3, \quad c = - \left(\frac{2}{\gamma} \right)^3 R - \frac{2}{\gamma^2} \sigma_\alpha^2 + \frac{\sigma_\alpha^2}{\gamma},$$

$$R = \ln k \sigma_\alpha^2, \quad A = \frac{4}{\gamma^2 \sigma_\alpha^2} (R+1) + \frac{1}{\gamma} - \frac{3}{2},$$

$$C = -\frac{2}{\gamma^2 \sigma_\alpha^2} R^2 + \left(\frac{1}{2} - \frac{1}{\gamma} - \frac{4}{\gamma^2 \sigma_\alpha^2} \right) R - \frac{1}{\gamma} + \frac{1}{2} + \ln \frac{4\Gamma\left(\frac{1}{2}\right) c_0}{\sqrt{k} \gamma^2},$$

and

$$g(x) = \frac{2x+1}{\gamma} - \frac{1}{2}.$$

Proof. See the Appendix.

Remark. When the noise (ε_t) follows a Gaussian distribution, then $c_0 = \frac{1}{\sqrt{2\pi}}$, $\gamma = 2$, $k = \frac{1}{2}$ and we find the results of Breidt and Davis (1998, p. 667).

We now define $M_n = \max(X_1, X_2, \dots, X_n)$, $n \geq 2$, the sample maxima of the process (X_t) . In the following theorem, we investigate the asymptotic distribution of M_n .

Theorem 1. Let (X_t) be a process defined in (2.5). Assume that the density of ε_t is given by the expression (2.1). Assume also that $\rho_\alpha(h) \ln h \rightarrow 0$, as $h \rightarrow +\infty$, where $\rho_\alpha(h)$ denotes the autocorrelation function of (α_t) defined in (1.2) and $\gamma \geq \frac{1}{2}$. Then there exist normalizing constants $(a_n > 0)$ and (b_n) such that

$$\mathbb{P}\{a_n(M_n - b_n) \leq x\} \rightarrow \exp(-e^{-x}),$$

where

$$a_n = \sigma_\alpha (2 \ln n)^{\frac{1}{2}}, \quad d_n = (\ln n)^{\frac{1}{2}}$$

and

$$b_n = c_1 d_n + c_2 \ln d_n + c_3 + c_4 \frac{\ln d_n}{d_n} + \frac{c_5}{d_n}, \quad (2.7)$$

with

$$c_1 = (2\sigma_\alpha^2)^{\frac{1}{2}}, \quad c_2 = a, \quad c_3 = a \left[\frac{3}{2} \ln 2 + \ln \frac{2}{\gamma} - \frac{\ln \sigma_\alpha^2}{2} - 1 \right],$$

$$c_4 = \left(\frac{1}{\gamma} - \frac{3}{2} \right) \frac{\sigma_\alpha}{\sqrt{2}},$$

$$\begin{aligned} c_5 = & \frac{-1}{2(2\sigma_\alpha^2)^{\frac{1}{2}}} \left\{ a^2 + \left(1 - R + \frac{1}{2} \ln 2\sigma_\alpha^2 \right) \sigma_\alpha^2 (1 - a) \right. \\ & + \left(\frac{\gamma^2}{4} \left(\frac{1}{2} - \frac{1}{\gamma} \right) + 2 + 2\sigma_\alpha^2 \right) \left(\frac{1}{\gamma} - \frac{1}{2} \right) \\ & \left. + (3 - a) \sigma_\alpha^2 \ln a - 2\sigma_\alpha^2 \ln \left(\frac{a^2 c_0 \sqrt{\pi}}{\sqrt{k}} \right) + \sigma_\sigma^2 \ln(2\pi) \right\}. \end{aligned}$$

Proof. See the Appendix.

3. Behaviour of the distribution of the maxima of (X_t) using finite samples

In this section we study the behaviour of the sample maxima's distribution M_n for different versions of the process (α_t) . We now describe these versions.

First, we consider the iid stochastic volatility model driven by a GED noise and the process (α_t) is defined by

$$\alpha_t = Z_t, \quad \{Z_t\} \text{ is iid } \mathcal{N}(0, \sigma_\alpha^2). \quad (3.1)$$

Secondly, we study the first-order SARV model driven by a GED noise and the process (α_t) is defined by

$$\alpha_t = \phi \alpha_{t-1} + Z_t, \quad \{Z_t\} \text{ is iid } \mathcal{N}(0, (1 - \phi^2) \sigma_\alpha^2), \quad (3.2)$$

with $|\phi| < 1$.

Finally, we consider the long memory stochastic volatility model (LMSV), (see Breidt *et al.*, 1998) driven by a GED noise and the process (α_t) is defined by

$$(1 - B)^d \alpha_t = Z_t, \quad \{Z_t\} \text{ is iid } \mathcal{N}\left(0, \frac{\sigma_\alpha^2 \Gamma^2(1-d)}{\Gamma(1-2d)}\right) \quad (3.3)$$

where B is the backward shift operator and $|d| < \frac{1}{2}$.

In the sequel, we highlight the influence of the parameters on the limiting distribution. We also study the importance of the different assumptions made in Theorem 1.

3.1 Assumptions

The result stated in Theorem 1 requires the condition $\rho_\alpha(h) \ln h \rightarrow 0$ as $h \rightarrow \infty$. The model (3.1) corresponds to an iid process thus $\rho_\alpha(h) = 0$, $h \neq 0$. The model (3.2) is an AR(1) process, thus $\rho_\alpha(h)$ decreases with the rate of $|\phi|^{|h|}$. The model (3.3) is a FARIMA(0, d , 0) process, it is known that $\rho_\alpha(h) \sim C(d) h^{2d-1}$, as $h \rightarrow \infty$, where $C(d)$ is some constant which depends on d , thus the speed of convergence of $\rho_\alpha(h)$ towards 0 is very slow in the latter case.

3.2 Influence of the parameters

In order to compare our results with those of Breidt and Davis (1998), we consider in our simulations the same values they use for σ_α^2 ($\sigma_\alpha^2 = 2.3976, 0.6933, 0.0953$), for ϕ ($\phi = 0.95$) and for d ($d = 0.4$). In Figures 1–4, the solid lines represent the empirical distribution functions for

1000 normalized maxima $\alpha_n(M_n - b_n)$ obtained from 1000 replications of samples size 1000. The dotted lines represent the limiting double exponential distribution. The quality of the convergence depends on σ_α and a_n (the coefficient a_n is given by $a_n = \sigma_\alpha^{-1} (2 \ln n)^{\frac{1}{2}}$). The influence of the tail-thickness parameter γ of the GED is also shown in Figures 1–4. We present the limiting distribution, when $\gamma = 1$ (Laplace distribution which has a thicker tail than the Gaussian distribution) and $\gamma = 3$ (thinner tail than the Gaussian distribution). Figure 1 of Breidt and Davis (1998) presents the case of $\gamma = 2$. The asymptotic behaviour of the normalized maxima is closely related to the value of σ_α as shown in Figures 1–4. We also study in Figures 3–4 the sensitivity of the convergence obtained in (2.7), with respect to the parameters ϕ and d , respectively, when (α_t) follows the models (3.2)–(3.3). We now detail these results.

3.2.1 GED noise with $\gamma = 3$

We now turn to the case of the stochastic volatility model (2.5) driven by a noise (ε_t) , characterized by tail-thickness parameter $\gamma = 3$, see Figure 1: the solid lines represent the empirical distribution function of the normalized maxima $a_n(M_n - b_n)$, with $n = 1000$ and the dotted lines represent the double exponential distribution. Looking at the nine graphs of Figure 1, we see that, for finite samples, the approximation of the empirical distribution function by the double exponential distribution is the best when (α_t) follows the model (3.2) with $\gamma = 0.95$ and $\sigma_\alpha^2 = 2.3976$.

3.2.2 Laplace noise $\gamma = 1$

We now abstract from the thinner tailed case and consider a stochastic volatility model driven with a Laplace noise $\gamma = 1$. The result is given in Figure 2. By reading across the rows of this picture, the panels again show the influence

of the value of the parameter σ_α^2 . The greater σ_α^2 , the better the rate of convergence of the normalized maxima of (X_t) to the double exponential distribution is. The dependence structure in (α_t) does not seem to have a great influence on that convergence. A comparison between Figure 1 of Breidt and Davis (1998) and Figure 2 shows clearly that the convergence of the normalized maxima is better when the driving noise (ε_t) is Gaussian. Despite the scale change in the x -axis of Figure 2, the comparison between Figure 1 and Figure 2, corresponding respectively to $\gamma = 3$ and $\gamma = 1$, reveals a better approximation when $\gamma = 3$. We also note that as γ increases, the mass of the empirical distribution shifts to the right.

3.2.3 Influence of the autoregressive parameter ϕ

Here we assume for simplicity that the noise (ε_t) follows a Gaussian law and we investigate the influence of the autoregressive parameter ϕ on the asymptotic behaviour of $a_n(M_n - b_n)$, when the process (α_t) follows the model (3.2). To determine the sensitivity of the results to this parameter, different values for ϕ have been considered: $\phi = 0.2$, $\phi = 0.95$ and $\phi = 0.99$. The results are given in Figure 3. By looking across the rows of Figure 3, we note that the smaller ϕ , the better the rate of convergence is. When the autoregressive parameter ϕ is equal to 0.99, the approximation is dramatically poor regardless of the value of σ_α , as shown in the last column of Figure 3. A possible explanation is the fact that the underlying process tends to become nonstationary when the parameter ϕ is close to 1. It seems that this situation needs other investigations. This case has already been partially examined by Horowitz (1980), Ballerini and McCormick (1989) and Niu (1997).

3.2.4 Importance of the long memory parameter d

Figure 4 shows the influence of the long memory parameter d on the convergence of the normalized maxima of the process (X_t) . For the sake of conciseness, we assume that the noise (ε_t) follows a Gaussian distribution. We compare the empirical distribution function of $a_n(M_n - b_n)$ with the double exponential distribution, when the process (X_t) follows the LMSV model with $d = 0.1$, $d = 0.2$ and $d = 0.4$. The smaller d , the better the rate of convergence (2.7) is. However, we have approximately the same behaviour as in the other cases when σ_α is equal to 0.0953.

3.2.5 Tail comparison

Here we propose a brief comparison between the tail behaviour of the two components of the process (X_t) defined in (2.5). When the process (ε_t) is GED, the density of the r.v. ς_t defined in (2.4) is given by

$$h(x) = c_0 \exp\left(\frac{x}{2} - ke^{\frac{x}{2}}\right).$$

Asymptotically the log-squared iid noise has an upper tail that is dominated by the Gaussian tail of the log-volatility term. The tail of (ς_t) tends asymptotically to

$$\frac{2c_0}{\gamma k} \exp\left(\frac{x}{2}(1 - \gamma)\right) \exp(-ke^{\frac{x}{2}}),$$

which tends toward zero faster than the Gaussian tail $\frac{1}{x} \exp\left(-\frac{x^2}{2}\right)$. Figure 5 shows the distribution's tails of the Gaussian linear process (α_t) defined in (1.2) with the noise (ς_t) defined in (2.4). The three panels correspond respectively to $\sigma_\alpha^2 = 2.3976$, $\sigma_\alpha^2 = 0.6933$ and $\sigma_\alpha^2 = 0.0953$. When $\sigma_\alpha^2 = 2.3976$, the Gaussian upper tail dominates the tail of ς_t for $\gamma = 1, 2, 3$, which is expected. However, in finite samples, when $\sigma_\alpha^2 = 0.0953$, the Gaussian upper tail is dominated by the upper tail of ς_t . Finally when

$\sigma_\alpha^2 = 0.6933$, all the tails tend to have the same behaviour as the Gaussian upper tail. Hence, in finite samples, the Gaussian tail dominance is not evident. This is why the double exponential approximation does not give good results with finite samples.

4. Conclusion

In this paper, we get the maxima's asymptotic distribution of a log-transformation for a process (Y_t) driven by a stochastic volatility model. We study the empirical behaviour of its asymptotic distribution. We also point out the influence of the tail-thickness parameter γ of the driving noise (ε_t) . Our findings show that the choice of γ , σ_α^2 , and the dependence structure of (α_t) affect the goodness-of-fit of the limiting distribution.

In practice, the results stated in Theorem 1, allow us to calculate quantile risk measures using the maximum block method, see for instance Embrechts *et al.* (1997). However, with finite samples care must be taken because of the approximation's lack of accuracy. This is one of the limitations to evaluate, for instance, the Value at Risk for real data, using this part of the extreme value theory. It would be interesting to investigate the asymptotic behaviour for more complicated non-stationary models. This will be done in a companion paper.

5. Appendix

Proof of Proposition 1. We focus only on the important steps of the proof. Breidt and Davis (1998) use a Tauberian argument in the Gaussian case to express the asymptotic approximation of the tail distribution of (X_t) . We use it again. In step 1, we give some expansions of the two derivatives $m(\lambda)$ and $S(\lambda)$ of the log-moment generating function of (X_t) defined in (2.5). In the

second step, we define an inverse function $m^{-1}(\cdot)$ of $m(\lambda)$ and we justify the inverse notation. Some useful expansions of functions in terms of $m^{-1}(x)$ are also provided. In the third step, we establish the asymptotic normality for the normalized Esscher transform of F which permits us to prove (2.6).

Step 1:

We first give some expansions of the two first derivatives of the log-moment generating function of the process (X_t) .

The log of the moment generating function of (ζ_t) is given by

$$\ln E \exp(\lambda \zeta_t) = \ln 2c_0 - \ln \gamma - \frac{2\lambda + 1}{\gamma} \ln k + \ln \Gamma\left(\frac{2\lambda + 1}{\gamma}\right). \quad (5.1)$$

Thus, the log of the moment-generating function of (X_t) is equal to

$$\begin{aligned} \ln C(\lambda) &= \frac{\lambda^2 \sigma_\alpha^2}{2} + \ln 2c_0 - \ln \gamma - \frac{2\lambda + 1}{\gamma} \ln k + \ln \Gamma\left(\frac{2\lambda + 1}{\gamma}\right) \\ &= \frac{\lambda^2 \sigma_\alpha^2}{2} - g(\lambda) \ln k - g(\lambda) + g(\lambda) \ln g(\lambda) + \ln \frac{2\sqrt{2}\Gamma\left(\frac{1}{2}\right)c_0}{\sqrt{k\gamma}} \\ &\quad + \mathcal{O}\left(\frac{1}{\lambda}\right) \end{aligned} \quad (5.2)$$

where

$$g(\lambda) = \frac{2\lambda + 1}{\gamma} - \frac{1}{2}.$$

The first two derivatives of $\ln C(\lambda)$ are

$$\begin{aligned} m(\lambda) &= \frac{d}{d\lambda} \ln C(\lambda) = \lambda \sigma_\alpha^2 - \frac{2}{\gamma} \ln k + \frac{2}{\gamma} \Psi\left(\frac{2\lambda + 1}{\gamma}\right) \\ &= \lambda \sigma_\alpha^2 - \frac{2}{\gamma} \ln k + \frac{2}{\gamma} \ln g(\lambda) + \mathcal{O}\left(\frac{1}{\lambda^2}\right) \end{aligned} \quad (5.3)$$

and

$$\begin{aligned} S^2(\lambda) &= \frac{d^2}{d\lambda^2} \ln C(\lambda) = \sigma_\alpha^2 + \left(\frac{2}{\gamma}\right)^2 \Psi' \left(\frac{2\lambda+1}{\gamma} \right) \\ &= \sigma_\alpha^2 + \mathcal{O} \left(\frac{1}{\lambda} \right), \end{aligned} \quad (5.4)$$

where $\Psi(\cdot)$ and $\Psi'(\cdot)$ are the digamma and trigamma functions respectively.

Step 2.

We set

$$m^{-1}(x) = \frac{\gamma g(x)}{2\sigma_\alpha^2} - \frac{a \ln g(x)}{\sigma_\alpha^2} + \frac{b \ln g(x)}{\sigma_\alpha^2(x)} + \frac{c}{\sigma_\alpha^2 g(x)} - \frac{d}{\sigma_\alpha^2}, \quad (5.5)$$

where

$$d = -\frac{2}{\gamma} \ln(k\sigma_\alpha^2).$$

It follows that

$$\ln g(m^{-1}(x)) = \ln \left(\frac{g(x)}{\sigma_\alpha^2} \right) - \frac{2a \ln g(x)}{g(x)} + \frac{D_1}{\gamma g(x)} + \mathcal{O} \left(\frac{\ln^2 g(x)}{g(x)^2} \right), \quad (5.6)$$

where

$$D_1 = -2d + \sigma_\alpha^2 - \frac{\gamma \sigma_\alpha^2}{2}.$$

After some computations, using (5.3), (5.5) and (5.6), we get

$$m(m^{-1}(x)) = x + \mathcal{O} \left(\frac{\ln^2 g(x)}{g(x)^2} \right) \quad (5.7)$$

which justifies the inverse notation.

Step 3.

We establish the asymptotic normality for the normalized Esscher transform of the distribution F (see Feigin and Yashchin, 1983). We write

$$\overline{F}(m(\lambda)) \sim \frac{\exp(-\lambda m(\lambda)) C(\lambda)}{(2\pi)^{\frac{1}{2}} \lambda S(\lambda)}. \quad (5.8)$$

Using the inverse of m defined in (5.5), we have

$$\overline{F}(x) \sim \frac{\exp(-xm^{-1}(x)) C(m^{-1}(x))}{(2\pi)^{\frac{1}{2}} m^{-1}(x) S(m^{-1}(x))}. \quad (5.9)$$

The remainder of the proof needs the following expansions:

$$\begin{aligned} \frac{\sigma_\alpha^2}{2} (m^{-1}(x))^2 &= \frac{\gamma^2 g(x)^2}{8\sigma_\alpha^2} + \frac{a^2 \ln^2 g(x)}{2\sigma_\alpha^2} - \frac{\gamma a g(x) \ln g(x)}{2\sigma_\alpha^2} \\ &\quad + \frac{(\gamma b + 2ad) \ln g(x)}{2\sigma_\alpha^2} \\ &\quad - \frac{\gamma d g(x)}{2\sigma_\alpha^2} + \frac{d^2 + \gamma c}{2\sigma_\alpha^2} + \mathcal{O}\left(\frac{\ln^2 g(x)}{g(x)^2}\right), \end{aligned}$$

and

$$\begin{aligned} g(m^{-1}(x)) \ln g(m^{-1}(x)) &= -\frac{\ln(\sigma_\alpha^2)}{\sigma_\alpha^2} g(x) + D_2 \ln g(x) \\ &\quad + \frac{g(x) \ln(g(x))}{\sigma_\alpha^2} - \frac{2a \ln^2 g(x)}{\gamma \sigma_\alpha^2} \\ &\quad + \frac{D_1}{\gamma \sigma_\alpha^2} (1 - \ln \sigma_\alpha^2) \\ &\quad + \mathcal{O}\left(\frac{\ln^2 g(x)}{g(x)}\right), \end{aligned}$$

where

$$D_2 = \frac{-ad + a^2 \ln \sigma_\alpha^2 - 2a}{\sigma_\alpha^2}.$$

We conclude the proof by using (5.4), (5.5), (5.7), (5.9).

Proof of Theorem 1. Using proposition 1, we get

$$F^n(u_n) \rightarrow \exp(-e^{-x}), \quad x \in \mathbb{R} \quad (5.10)$$

where $u_n = a_n^{-1}x + b_n$. Now we need to show that

$$|P(M_n \leq u_n) - F^n(u_n)| \rightarrow 0 \quad \text{as } n \rightarrow +\infty. \quad (5.11)$$

The proof of (5.11) uses the Normal comparison lemma, see Leadbetter *et al.* (1983), page 81, and follows the lines of Breidt and Davis (1998). First, we establish

$$|P(M_n \leq u_n) - F^n(u_n)| \leq nK \sum_{i=1}^n |\rho_\alpha(i)| \left(E \exp \left\{ -\frac{(u_n - \varsigma_1)^2}{2\sigma_\alpha^2(1 + |\rho_\alpha(i)|)} \right\} \right)^2 \quad (5.12)$$

where K is a positive constant, $\rho_\alpha(\cdot)$ the autocorrelation function of the process (α_t) and $\varsigma_t = \ln \varepsilon_t^2$. It is easy to show that for $\gamma \geq \frac{1}{2}$, we have

$$\overline{F}_\varepsilon(x) \approx \frac{c_0}{\gamma k} x^{1-\gamma} e^{-kx^\gamma} \quad \text{as } x \rightarrow +\infty.$$

Using the asymptotic relationship for the tail probability of the GED, it can be shown that

$$\begin{aligned} E \exp \left\{ -\frac{(u_n - \varsigma_1)^2}{2\sigma_\alpha^2(1 + |\rho_\alpha(i)|)} \right\} &\leq K' (\ln n)^{\left[\frac{1}{2(1 + |\rho_\alpha(i)|)} \right]} \\ &\quad E \left(\overline{F}_\varepsilon^{\frac{1}{1 + |\rho_\alpha(i)|}} (\sigma_\alpha^{-1}(u_n - \varsigma_1)) \right) \\ &\quad + o(n^{-1}), \end{aligned} \quad (5.13)$$

where $[\cdot]$ stands for the integer part and K' a positive constant. By Jensen's inequality and the relation (5.10), we have

$$\left(E \exp \left\{ -\frac{(u_n - \varsigma_1)^2}{2\sigma_\alpha^2(1 + |\rho_\alpha(i)|)} \right\} \right)^2 \leq K'' (\ln n)^{\frac{1}{1 + |\rho_\alpha(i)|}} n^{\frac{-2}{1 + |\rho_\alpha(i)|}} \quad (5.14)$$

where K'' is a positive constant. The remainder of the proof is similar to the one of Breidt and Davis (1998), page 671; see also Leadbetter *et al.* (1983), page 86.

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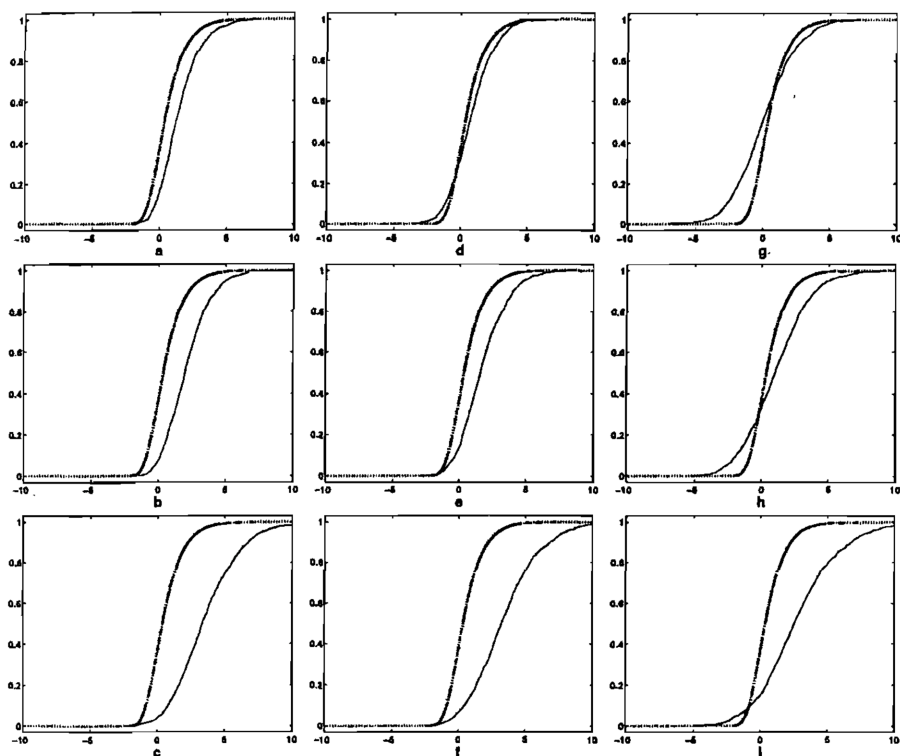


Figure 1. Comparison between the empirical distribution function and the double exponential distribution for the process (X_t) defined in (2.5) with a GED driving noise (ε_t) with $\gamma = 3$. In first line, for the three graphs, $\sigma_\alpha^2 = 2.3976$, in the second line, $\sigma_\alpha^2 = 0.6933$ and in the third line, $\sigma_\alpha^2 = 0.0953$. a, b, c: $\alpha_t \sim IID$; d, e, f: $\alpha_t \sim AR(1)$, $\phi = 0.95$; g, h, i: $\alpha_t \sim FARIMA(0, d, 0)$, $d = 0.4$.

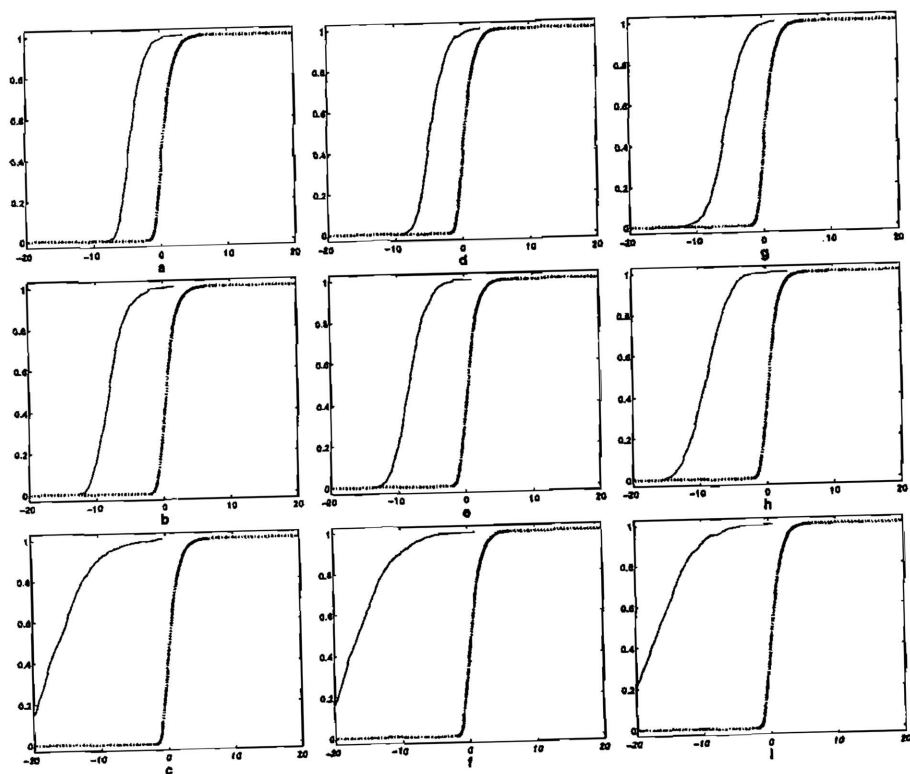


Figure 2. Comparison between the empirical distribution function (solid lines) for 1000 normalized maxima of (X_t) defined in (2.5) and the double exponential distribution (dotted lines) with a GED driving noise (ε_t) with $\gamma = 1$. In first line, for the three graphs, $\sigma_\alpha^2 = 2.3976$, in the second line, $\sigma_\alpha^2 = 0.6933$ and in the third line, $\sigma_\alpha^2 = 0.0953$. a, b, c: $\alpha_t \sim IID$; d, e, f: $\alpha_t \sim AR(1)$, $\phi = 0.95$; g, h, i: $\alpha_t \sim FARIMA(0, d, 0)$, $d = 0.4$.

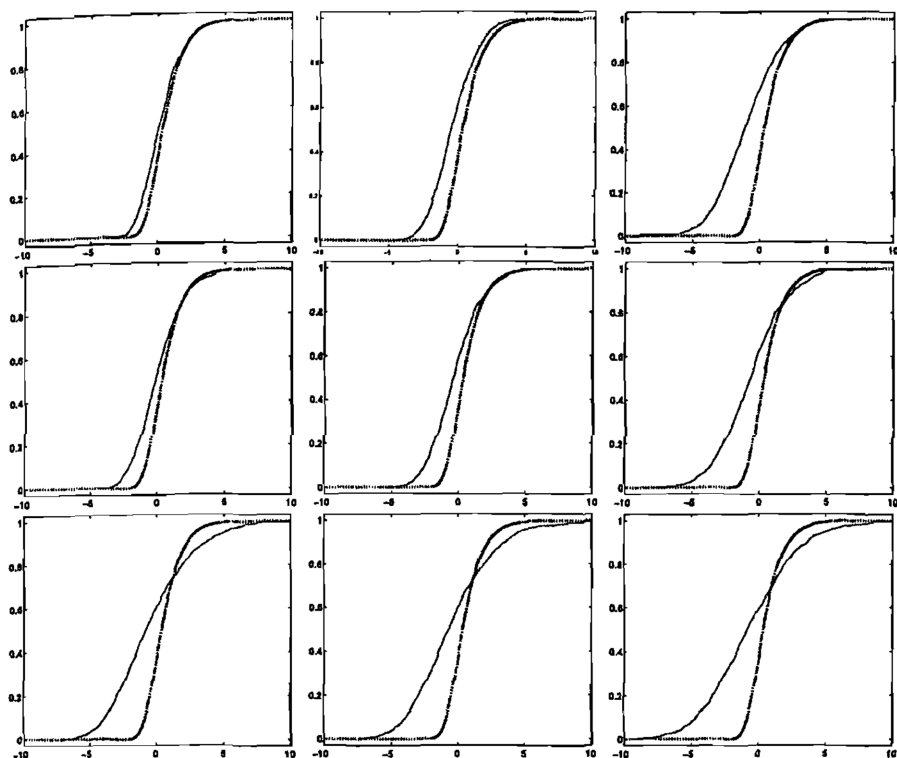


Figure 3. Comparison between the empirical distribution function (solid lines) for 1000 normalized maxima of (X_t) defined in (2.5) and the double exponential distribution (dotted lines), for ARSV model, when the driving noise (ε_t) is Gaussian. For each line, σ_α^2 is constant. The values of σ_α^2 are the same as in Figure 1. The columns correspond, respectively, to $\phi = 0.2$, $\phi = 0.95$ and $\phi = 0.99$.

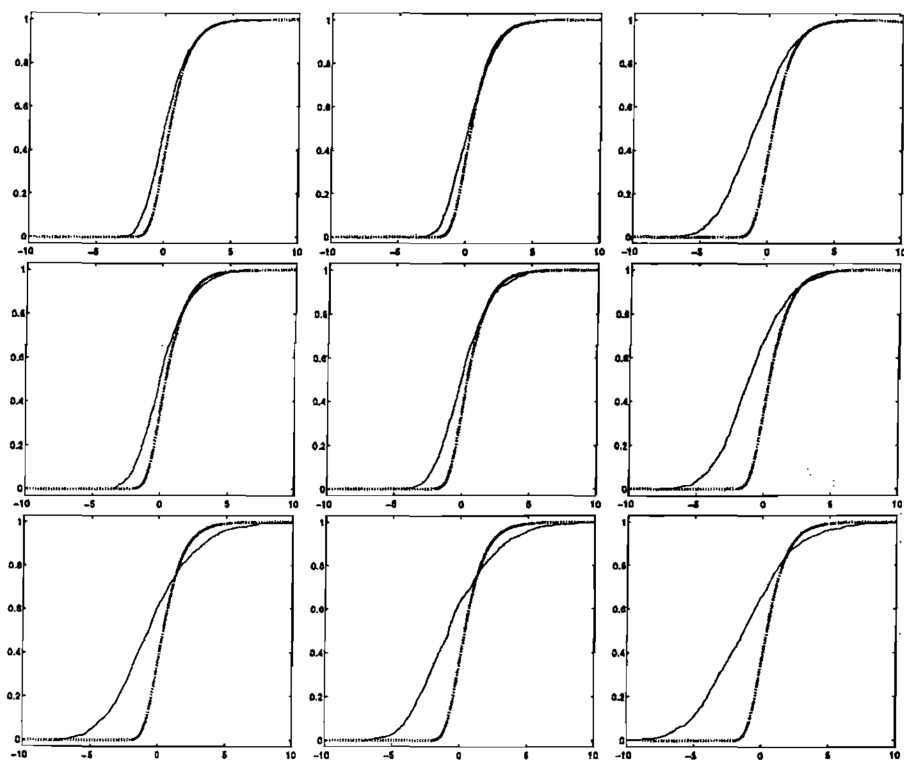


Figure 4. Comparison between the empirical distribution function (solid lines) for 1000 normalized maxima of (X_t) defined in (2.5) and the double exponential distribution (dotted lines), for LMSV model, when the driving noise (ε_t) is Gaussian. For each line, σ_α^2 is constant. The values of σ_α^2 are the same as in Figure 1. The columns correspond, respectively, to $d = 0.1$, $d = 0.2$ and $d = 0.4$.

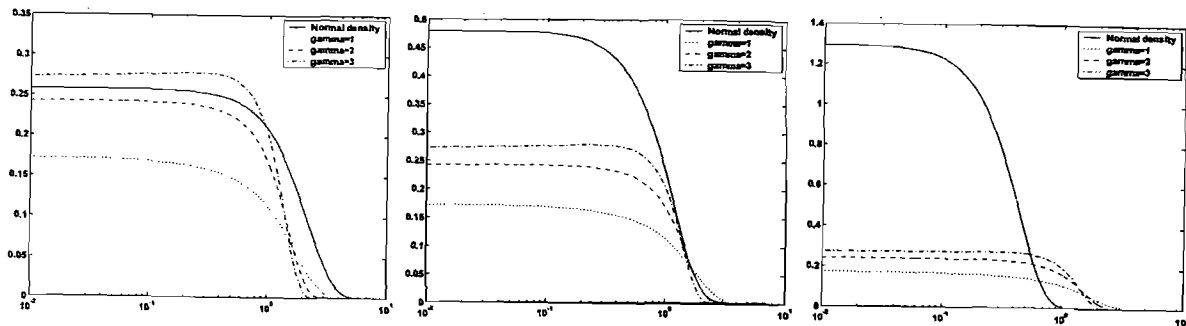


Figure 5. Comparison between the densities of α_t defined in (1.2) (solid line) and ζ_t defined in (2.4) for different values of γ ($\gamma = 1, 2, 3$). The three graphs correspond respectively to $\sigma_\alpha^2 = 2.3976$, $\sigma_\alpha^2 = 0.6933$ and $\sigma_\alpha^2 = 0.0953$. We choose a log-scale on the x-axis.